GAME CHANGERS

Yann LeCun bet against the room twice

He was right once. Now he's betting a billion dollars he's right again.

By **Vyom Ramani** | editor@digit.in

Most people in AI have one contrarian bet in them, the idea they defend past the point of embarrassment until either the field catches up or their career quietly ends. Yann LeCun has made this bet twice, decades apart, against two completely different rooms. The first time, the room was 1990s AI, dismissive of neural networks as a dead end, and LeCun spent a decade proving them wrong with pattern recognition nobody outside a handful of labs cared about. The second time, the room is the one he helped build, an industry that bet the next trillion dollars on large language models, and LeCun spent the last four years standing in the middle of it insisting the entire premise is a detour. He was Meta's chief AI scientist while saying this. Then he left to prove it with his own money and, more importantly, with $1.03 billion of other people's.

> "The analogy I've been using is the fact that perhaps an equivalent event in the history of humanity to what might be provided by Generalization of AI assistant is the invention of the printing press. It made everybody smarter."

The heresy that worked

LeCun trained under Geoffrey Hinton in the 1980s, at a moment when neural networks were considered a fringe pursuit that had already failed once and would probably fail again. At Bell Labs through the 1990s, he built convolutional neural networks, LeNet, a system that could read handwritten digits well enough that it ended up processing a meaningful share of the checks deposited in American banks. That's not a research footnote. It was a working commercial deployment of deep learning a full two decades before the term entered normal conversation, at a time when most of the field had moved on to support vector machines and decided the neural network idea had run its course. LeCun didn't move on. He kept refining the architecture through what the industry still calls the AI winter, years when funding dried up and publishing on neural nets was a way to make a paper harder to get accepted, not easier.

The vindication took until 2012, when a neural network trained on ImageNet blew every existing computer vision benchmark out of the water and the architecture underneath it traced straight back to what LeCun had been building at